B1
Potential problems:
Basking space
feeding
entertainment

B1 Documentation of plausible prior attempts to solve the problem and/or related problems

Bulla, Andrew, et al. “The Effect of Preference Assessment-Informed Enrichment Device Colour on Biting and Foraging Behaviour in Loggerhead Sea Turtles.” Journal of Zoo and Aquarium Research, vol. 12, no. 4, 2024, pp. 232–242, www.jzar.org/jzar/article/view/801, https://doi.org/10.19227/jzar.v12i4.801.

This article experiments with different colored enrichment devices for turtles. It found that turtles do have color preferences, but when preferred food is added, color makes a minimal difference in how much they interact with them.

“Enriching the Lives of Rescued Sea Turtles - ARCHELON - The Sea Turtle Protection Society of Greece.” ARCHELON - The Sea Turtle Protection Society of Greece, 2026, https://archelon.gr/en/news/enriching-the-lives-of-rescued-sea-turtles-1. Accessed 2 Apr. 2026.

This article explores how environmental enrichment effects turtles well being. These stimulating environments were found to positively affect turtles well being. In the case of their experiment, they also used the environment to train the turtles on skills they would need for there release into the wild, such as diving.

Therrien, Corie L., et al. “Experimental Evaluation of Environmental Enrichment of Sea Turtles.” Zoo Biology, vol. 26, no. 5, 2007, pp. 407–416, https://doi.org/10.1002/zoo.20145. Accessed 2 Apr. 2026.

This article tested the swimming patterns of turtles with and without enrichment. It found that turtles with enrichment spent 69% (additive) less time resting and in pattern swimming, and significantly more time in focused behavior.

“Nocturnal Basking in Freshwater Turtles: A Global Assessment.” USGS, 2023, www.usgs.gov/publications/nocturnal-basking-freshwater-turtles-a-global-assessment. Accessed 2 Apr. 2026.

This article documented the basking times of turtles. They found that in certain areas, basking was done nocturnally, suggesting that the temperature of the environment plays a role, however they failed to identify the primary cause.

Tetzlaff, Sasha, et al. “Tradeoffs with Growth and Behavior for Captive Box Turtles Head-Started with Environmental Enrichment.” Diversity, vol. 11, no. 3, 7 Mar. 2019, p. 40, https://doi.org/10.3390/d11030040.

This article studied the relation to growth and enrichment. It found that turtles with more enrichment, actually grew slower than those without, despite the beneficial behaviors the enrichment provided.

B2 The analysis of past and current attempts to solve the problem

All studies done on the enrichment of turtles showed significant positive impacts. Most notably, it increased their wellbeing and decreased the amount of time spent resting and doing stereotypic swimming. While turtles do have preferences on certain enrichments, as long as it is within reason and food is involved, there should be no significant decrease in the affect the enrichment has on the turtles. The one drawback to enrichment is that It was shown to slow down growth, even though the rest of their physical and mental health showed to have a positive increase. The main draw back for slowed growth was in turtles that were planning on being released, as they would have a harder time surviving at a smaller size. In captivity, there should be no significant drawback for the slowed growth, and it could also be considered beneficial considering the size of the tank, and their limited space. There was also the study of basking for turtles. This suggests that an adjustment to the heat may be beneficial depending on things like room temperature and how often these specific turtles are basking.